PRACTICAL WORK 4

Multilayer Perceptron Learning

Dataset (see attached files)

- Adult Census Income Dataset
  - ❖ https://www.kaggle.com/datasets/priyamchoksi/adult-census-income-dataset
- Titanic Dataset
  - ❖ https://www.kaggle.com/datasets/brendan45774/test-file

Learning Objectives

- Understand the architecture of Multilayer Perceptrons
- Prepare data for neural network training
- Implement MLP models using Python libraries
- Experiment with hidden layers, activation functions, and learning parameters
- Evaluate model performance using classification metrics
- Analyze how network architecture influences model accuracy

Task

Complete the following task:

- missing values, duplicates, noise/outliers, inconsistencies
- Apply imputation: mean/ median/KNN
- Apply Normalization techniques (Minmax scaler; Z-score)
  - ❖ Compare distributions before and after scaling
- Standardize features using PCA
  - ❖ Plot cumulative explained variance
- Data Preparation for Neural Networks
  - ❖ Implementing a Multilayer Perceptron
  - ❖ Train multiple MLP models by modifying:
    - ✓ Number of hidden layers: (10, 20), 50)
    - ✓ Activation functions: (ReLU, Tanh, Logistic (Sigmoid)
    - ✓ Plot the training loss curve of the MLP model.
    - ✓ Compare: Training accuracy vs. Testing accuracy

Deliverables

- Write a summarized report describing each task results.
  - ❖ Include the screenshots of plots
- Submit the code